

PROPERTIES OF SEAWATER

In addition to the dependence on temperature and pressure, the physical properties of seawater vary with the concentration of the dissolved constituents. A convenient parameter for describing the composition is the salinity, S , which is defined in terms of the electrical conductivity of the seawater sample. The defining equation for the practical salinity is:

$$S = a_0 + a_1 K^{1/2} + a_2 K + a_3 K^{3/2} + a_4 K^2 + a_5 K^{5/2},$$

where K is the ratio of the conductivity of the seawater sample at 15°C and atmospheric pressure to the conductivity of a potassium chloride solution in which the mass fraction of KCl is 0.0324356, at the same temperature and pressure. The values of the coefficients are:

$$\begin{aligned} a_0 &= 0.0080 & a_3 &= 14.0941 \\ a_1 &= -0.1692 & a_4 &= -7.0261 \\ a_2 &= 25.3851 & a_5 &= 2.7081 \\ \Sigma a_i &= 35.0000 \end{aligned}$$

Thus when $K = 1$, $S = 35$ exactly (S is normally quoted in units of ‰, i.e., parts per thousand). The value of S can be roughly equated with the mass of dissolved material in grams per kilogram of seawater. Salinity values in the open oceans at mid latitudes typically fall between 34 and 36.

It is customary in oceanography to define the pressure at a given point as the pressure due to the column of water between that point and the surface. Thus by convention $P = 0$ at the sea surface. To a good approximation the pressure in decibars (dbar) can be equated to the depth in meters. Thus at 45° latitude the pressure is 5000 dbar at 4902 m, 10000 dbar at 9700 m.

The freezing point of seawater varies with salinity and pressure as follows (freezing point in °C):

P/dbar	$S = 0$	5	10	15	20	25	30	35	40
0	0.000	-0.274	-0.542	-0.812	-1.083	-1.358	-1.638	-1.922	-2.212
50	-0.038	-0.311	-0.580	-0.849	-1.121	-1.396	-1.676	-1.960	-2.250
100	-0.075	-0.349	-0.618	-0.887	-1.159	-1.434	-1.713	-1.998	-2.287
500	-0.377	-0.650	-0.919	-1.188	-1.460	-1.735	-2.014	-2.299	-2.589

The first table below gives several properties of seawater as a function of temperature for a salinity of 35. The second and third give density and electrical conductivity as a function of salinity at several temperatures, and the last lists typical concentrations of the main constituents of seawater as a function of salinity.

REFERENCES

1. *The Practical Salinity Scale 1978 and the International Equation of State of Seawater 1980*, Unesco Technical Papers in Marine Science No. 36, Unesco, Paris, 1981; sections No. 37, 38, 39, and 40 in this series give background papers and detailed tables.
2. Kennish, M. J., *CRC Practical Handbook of Marine Science*, CRC Press, Boca Raton, FL, 1989.
3. Poisson, A. *IEEE J. Ocean. Eng.* OE-5, 50, 1981.
4. Webster, F., in *AIP Physics Desk Reference*, E. R. Cohen, D. R. Lide and G. L. Trigg, eds., Springer-Verlag, New York, 2002.

Properties of Seawater as a Function of Temperature at Salinity $S = 35$ and Normal Atmospheric Pressure

ρ = density in g/cm³

$\beta = (1/\rho) (d\rho/dS)$ = fractional change in density per unit change in salinity

$\alpha = (1/\rho) (d\rho/dt)$ = fractional change in density per unit change in temperature (°C⁻¹)

κ = electrical conductivity in S/cm

η = viscosity in mPa s (equal to cP)

c_p = specific heat in J/kg °C

v = speed of sound in m/s

$t/^\circ\text{C}$	$\rho/\text{g cm}^{-3}$	$10^7\beta$	$10^7\alpha/^\circ\text{C}^{-1}$	$\kappa/\text{S cm}^{-1}$	$\eta/\text{mPa s}$	$c_p/\text{J kg}^{-1} ^\circ\text{C}^{-1}$	$v/\text{m s}^{-1}$
0	1.028106	7854	526	0.029048	1.892	3986.5	1449.1
5	1.027675	7717	1136	0.033468	1.610		
10	1.026952	7606	1668	0.038103	1.388	3986.3	1489.8
15	1.025973	7516	2141	0.042933	1.221		
20	1.024763	7444	2572	0.047934	1.085	3993.9	1521.5
25	1.023343	7385	2970	0.053088	0.966		
30	1.021729	7338	3341	0.058373	0.871	4000.7	1545.6
35	1.019934	7300	3687				
40		7270	4004			4003.5	1563.2

PROPERTIES OF SEAWATER (continued)

Density of Surface Seawater in g/cm³ as a Function of Temperature and Salinity

<i>t</i> /°C	<i>S</i> = 0	<i>S</i> = 5	<i>S</i> = 10	<i>S</i> = 15	<i>S</i> = 20	<i>S</i> = 25	<i>S</i> = 30	<i>S</i> = 35	<i>S</i> = 40
0	0.999843	1.003913	1.007955	1.011986	1.016014	1.020041	1.024072	1.028106	1.032147
5	0.999967	1.003949	1.007907	1.011858	1.015807	1.019758	1.023714	1.027675	1.031645
10	0.999702	1.003612	1.007501	1.011385	1.015269	1.019157	1.023051	1.026952	1.030862
15	0.999102	1.002952	1.006784	1.010613	1.014443	1.018279	1.022122	1.025973	1.029834
20	0.998206	1.002008	1.005793	1.009576	1.013362	1.017154	1.020954	1.024763	1.028583
25	0.997048	1.000809	1.004556	1.008301	1.012050	1.015806	1.019569	1.023343	1.027128
30	0.995651	0.999380	1.003095	1.006809	1.010527	1.014252	1.017985	1.021729	1.025483
35	0.994036	0.997740	1.001429	1.005118	1.008810	1.012509	1.016217	1.019934	1.023662
40	0.992220	0.995906	0.999575	1.003244	1.006915	1.010593	1.014278	1.017973	1.021679

Electrical Conductivity of Seawater in S/cm as a Function of Temperature and Salinity

<i>t</i> /°C	<i>S</i> = 5	<i>S</i> = 10	<i>S</i> = 15	<i>S</i> = 20	<i>S</i> = 25	<i>S</i> = 30	<i>S</i> = 35	<i>S</i> = 40
0	0.004808	0.009171	0.013357	0.017421	0.021385	0.025257	0.029048	0.032775
5	0.005570	0.010616	0.015441	0.020118	0.024674	0.029120	0.033468	0.037734
10	0.006370	0.012131	0.017627	0.022947	0.028123	0.033171	0.038103	0.042935
15	0.007204	0.013709	0.019905	0.025894	0.031716	0.037391	0.042933	0.048355
20	0.008068	0.015346	0.022267	0.028948	0.035438	0.041762	0.047934	0.053968
25	0.008960	0.017035	0.024703	0.032097	0.039276	0.046267	0.053088	0.059751
30	0.009877	0.018771	0.027204	0.035330	0.043213	0.050888	0.058373	0.065683

Composition of Seawater and Ionic Strength at Various Salinities (Ref. 2)

Constituent	Expressed as molality			As grams per kilogram of seawater		
	<i>S</i> = 30	<i>S</i> = 35	<i>S</i> = 40	<i>S</i> = 30	<i>S</i> = 35	<i>S</i> = 40
Cl ⁻	0.482	0.562	0.650	16.58	19.33	22.36
Br ⁻	0.00074	0.00087	0.00100	0.057	0.067	0.078
F ⁻		0.00007			0.001	
SO ₄ ²⁻	0.0104	0.0114	0.0122	0.97	1.06	1.14
HCO ₃ ⁻	0.00131	0.00143	0.00100	0.078	0.085	0.059
NaSO ₄ ⁻	0.0085	0.0108	0.0139	0.98	1.25	1.60
KSO ₄ ⁻	0.00010	0.00012	0.00015	0.013	0.016	0.020
Na ⁺	0.405	0.472	0.544	9.03	10.53	12.13
K ⁺	0.00892	0.01039	0.01200	0.338	0.394	0.455
Mg ²⁺	0.0413	0.0483	0.0561	0.974	1.139	1.323
Ca ²⁺	0.00131	0.00143	0.00154	0.051	0.056	0.060
Sr ²⁺	0.00008	0.00009	0.00011	0.007	0.008	0.009
MgHCO ₃ ⁺	0.00028	0.00036	0.00045	0.023	0.030	0.037
MgSO ₄	0.00498	0.00561	0.00614	0.582	0.655	0.717
CaSO ₄	0.00102	0.00115	0.00126	0.135	0.152	0.166
NaHCO ₃	0.00015	0.00020	0.00024	0.012	0.016	0.020
H ₃ BO ₃	0.00032	0.00037	0.00042	0.019	0.022	0.025
Ionic strength	0.5736	0.6675	0.7701			

Salinity

Salinity is the dissolved salt content of water. It is a general term used to describe the levels of different salts such as sodium chloride, magnesium and calcium sulfates, and bicarbonates.

Water salinity			
<u>Fresh water</u>	<u>Brackish water</u>	<u>Saline water</u>	<u>Brine</u>
< 0.05 %	0.05 – 3 %	3 – 5 %	> 5 %
< 500 ppm	500 – 30,000 ppm	30,000 – 50,000 ppm	> 50,000 ppm

Salinity measurement

There are two main methods of determining the salt content of water: Total Dissolved Salts (or Solids) and Electrical Conductivity.

Total Dissolved Salts (TDS) is measured by evaporating a known volume of water to dryness, then weighing the solid residue remaining. Electrical conductivity (EC) is measured by passing an electric current between two metal plates (electrodes) in the water sample and measuring how readily current flows (ie conducted) between the plates. The more dissolved salt in the water, the stronger the current flow and the higher the EC. Measurements of EC can be used to give an estimate of TDS.

Measurement of TDS is tedious and cannot be carried out in the field. EC measurement is much quicker and simpler and is very useful for field measurement. There are however a few simple precautions to note in doing so and these are outlined here.

Salinity units

TDS is recorded in milligrams of dissolved solid in one litre of water (mg/L). Parts per million (ppm) is equivalent to mg/L but it is not a favored unit. EC measures the charge carrying ability (ie conductance) of liquid in a measuring cell of specific dimensions. It is necessary to clearly define the units of both conductance and length when talking ECs. To say a water sample is 2000 EC, is like saying a table is 2000 long, without specifying millimetres, centimetres or metres. The standard EC unit used by the Victorian Salinity Program and the Murray Darling Basin Commission is microSiemens per centimetre ($\mu\text{S}/\text{cm}$) at 25°C.

You will however see other units and need to be aware of the relationships between them. $\mu\text{S}/\text{cm}$ relates to other units as follows:

$$\begin{aligned}1000 \mu\text{S}/\text{cm} &= 1 \text{ deciSiemen/metre (dS/m)} \\1000 \mu\text{S}/\text{cm} &= 1 \text{ milliSiemen/centimetre (mS/cm)} \\10 \mu\text{S}/\text{cm} &= 1 \text{ milliSiemen/metre (mS/m)}\end{aligned}$$

Relationship of total dissolved salts to EC

EC can be effectively converted to TDS for natural Victorian waters by the following relationship:

$$\text{TDS (mg/L)} = \text{EC } (\mu\text{S}/\text{cm at } 25^\circ\text{C}) \times 0.6$$

Parts-per notation is used, especially in science and engineering, to denote relative proportions in measured quantities; particularly in low-value (high-ratio) proportions at the parts-per-million (ppm) 10^{-6} , parts-per-billion (ppb) 10^{-9} , and parts-per-trillion (ppt) 10^{-12} level. Since parts-per notations are quantity-per-quantity measures, they are known as dimensionless quantities.

The TDS in water can be measured in:

milligrams per litre (mg/L)
parts per million (ppm).